



GS FOUNDATION 2024-25

GEOGRAPHY - 9

Water in the Atmosphere - III

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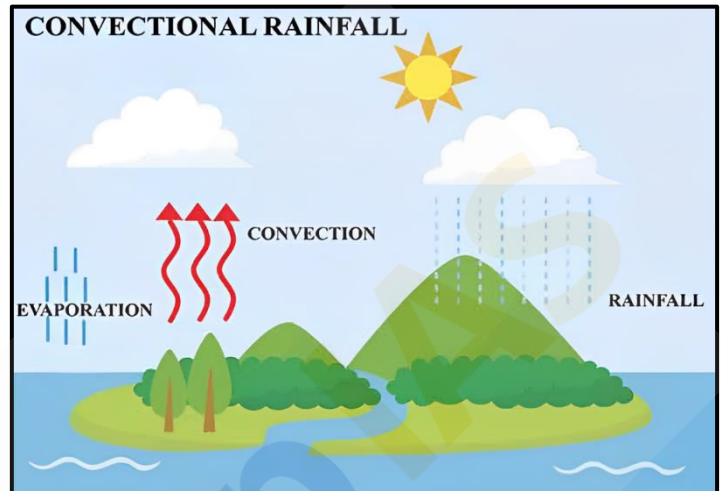
Precipitation

Types of Precipitation:

All precipitation occurs from clouds, and by far the most important cause of clouds is the adiabatic cooling resulting from the upward movement of air. Therefore, precipitation is classified on the basis of the conditions under which large masses of moist air are actually induced to rise to higher elevations.

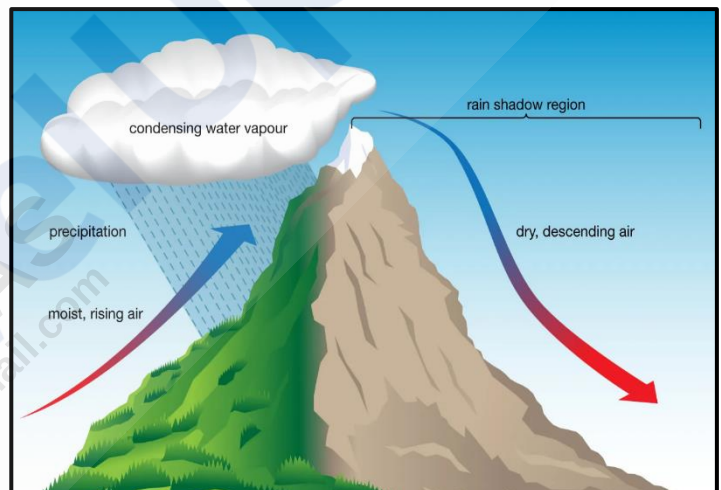
Convictional Precipitation:

Excessive heating of the earth's surface in tropical region results in the **vertical air currents**. These currents, lift the warm moist air to higher strata of atmosphere. When the temperature of such a humid air starts falling below dew point continuously, clouds are formed. These clouds cause heavy rainfall which is associated with **lightning and thunder**. This type of rainfall is called **conventional rainfall**. It is very common in **equatorial region** where it is a daily phenomenon in the afternoon.



Orographic or Relief Precipitation:

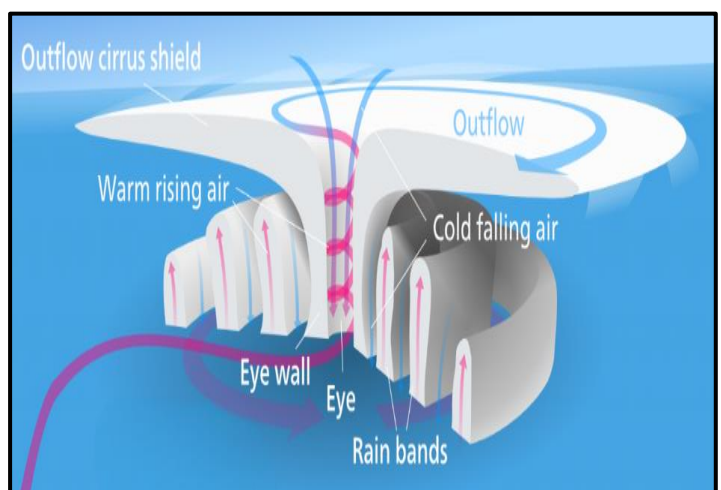
Orographic rainfall is formed where **air rises and cools because of a topographic barrier**. When their temperature falls below dew point, clouds are formed. These clouds cause **widespread rain on the windward slopes** of the mountain range. This type of rain is called **orographic rainfall**. However, when these winds cross over the mountain range and descend along the leeward slopes, they get warm and cause little rain. Region lying on the **leeward side of the mountain receiving little rain** is called **rain shadow area**. A famous example of orographic rainfall is **Cherrapunji** on the southern margin of the **Khasi Hills in Meghalaya**.



Cyclonic or Frontal Precipitation:

Cyclonic Precipitation in Tropical Regions:

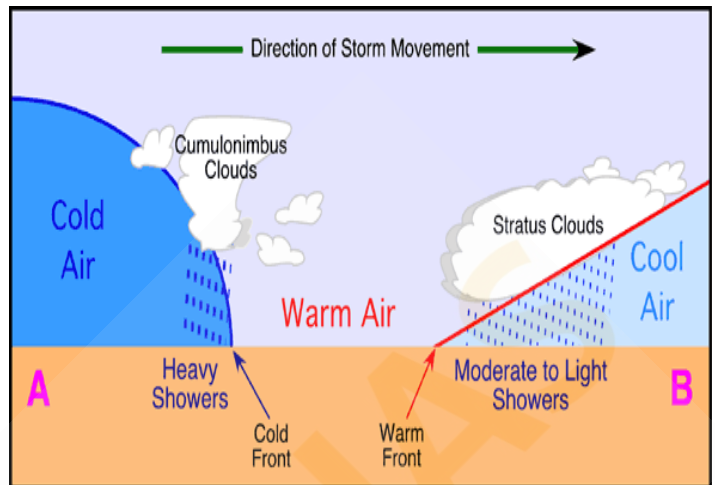
- In **tropical regions**, cyclonic precipitation occurs when air currents with different temperatures and moisture levels meet at an angle. The **warmer, moist air is forced to rise over the cooler, denser air, which remains lower**. Additionally, when air masses from different directions converge towards a center, as in cyclonic circulation, some air is pushed upward. This convergence leads to cloud formation and precipitation.



- In tropical areas, where there is little difference in temperature and humidity between the converging air masses, the lifting is almost vertical and typically accompanied by convection. In these conditions, **convergence initiates the upward movement** of convectively unstable air masses, resulting in the formation of **cumulonimbus clouds and heavy showers**.

Frontal Precipitation in Temperate Regions:

- In **temperate regions**, the boundaries where warm and cold air masses meet are known as **fronts**. Frontal precipitation occurs when warm, moist air is gradually lifted above the cold air wedge at these fronts. In stable air, this convergence typically leads to the formation of stratiform clouds, resulting in a grey overcast sky and prolonged, steady precipitation. The primary cause of frontal precipitation is the **mixing of air along the fronts**.
- When precipitation occurs along warm fronts, it usually takes the form of **drizzle**. This type of precipitation is widespread and lasts for an extended period. In contrast, precipitation associated with cold fronts often manifests as **short-lived thunderstorms**.



Distribution of Precipitation:

The distribution of precipitation across the globe is far from uniform, influenced by a combination of global pressure and wind belts, the distribution of land and water bodies, and the nature of relief features.

The **global average annual precipitation** stands at approximately **97.5 centimeters**. However, **land areas receive less precipitation compared to the oceans**, contributing to significant regional variations.

Regional Variations in Precipitation:

Regions of Heavy Precipitation:

- Regions receiving **over 200 centimeters** of annual precipitation fall into this category. These areas include the **equatorial coastal regions of the tropical zone** and the **west-coastal regions of the temperate zone**.

Regions of Moderate Precipitation:

- This category includes regions that receive **between 100 and 200 centimeters** of annual precipitation. These areas are typically adjacent to regions of heavy precipitation and include the **eastern coastal regions of the subtropical zone** and the **coastal regions of the warm temperate zone**.

Regions of Less Precipitation:

- Regions receiving **between 50 and 100 centimeters** of annual precipitation fall into this category. These areas are usually located in the **interior parts of the tropical zone** and the **eastern interior parts of the temperate zone**.

Regions of Scanty Precipitation:

- Areas that receive **less than 50 centimeters** of annual precipitation are categorized as regions of scanty precipitation. These regions include the **rain shadow (leeward) sides of mountain ranges, interior parts of continents, western margins of continents along the tropics, and high latitudes**. This category encompasses the tropical, temperate, and cold deserts of the world.

Seasonal Variations in Precipitation:

The regional variations in the distribution of precipitation across different parts of the world are often assessed based on average annual precipitation. However, this approach fails to accurately reflect the nature of precipitation, particularly in **regions where seasonal fluctuations are common, such as arid, semi-arid, or sub-humid areas**. Therefore, studying the seasonal variations in precipitation is crucial for a comprehensive understanding.

Equatorial and Temperate Regions:

- **Equatorial regions** and the **western parts** of temperate lands receive **precipitation throughout the year**. In equatorial regions, precipitation typically occurs in the form of **conventional rain**, while in **temperate regions**, it is mainly **cyclonic or orographic**, driven by westerly winds.

Winter Precipitation:

- Approximately 2 percent of the world's land areas receive **precipitation exclusively in winter**. This includes the **Mediterranean regions** and the **Coromandel Coast of India**. The seasonal shift in pressure and planetary wind systems causes these regions to experience **dry summers**, as they fall under the influence of sub-tropical high-pressure belts and dry trade winds when reaching the western margins of continents.

Summer Precipitation:

- The remaining parts of the world receive **precipitation predominantly in summer**. This highlights the significant seasonal variation in precipitation **experienced by most regions globally**. The seasonal distribution of precipitation is crucial for assessing its effectiveness. For instance, scanty precipitation during the short growing season in high latitudes is more effective than heavy precipitation in lower latitudes. Similarly, precipitation in the form of dew, fog, and mist in regions like **Central India** and the **Kalahari Desert** has a substantial impact on standing crops and natural vegetation.

Factors Affecting Distribution of Precipitation:

- **Moisture Supply** - The main determinant of rainfall in any area is the supply of moisture to the atmosphere. Equatorial and tropical regions experience the highest levels of evaporation, resulting in abundant moisture and consequently, significant rainfall. In contrast, coastal areas have more moisture compared to the interior parts of continents. Frigid regions, characterized by low evaporation rates, receive scant precipitation.
- **Wind Direction** - The direction of winds, particularly within the belts of trade winds and westerlies, plays a crucial role in rainfall distribution. Winds blowing from the sea towards the land typically bring rainfall, while those blowing from the land are generally dry. Winds moving from higher to lower latitudes tend to heat up and do not produce rain, whereas winds traveling from lower to higher latitudes cool down and often cause rainfall. Sub-tropical deserts experience minimal rainfall due to the presence of off-shore winds.
- **Ocean Currents** - Ocean currents also impact rainfall patterns. Warm currents are usually associated with warm, moist winds that bring rainfall, while cold currents are linked to cold, dry winds that result in little to no precipitation.
- **Topography** - The presence of mountains across the path of prevailing winds can significantly affect rainfall. The windward side of a mountain typically receives more rainfall, while the leeward side experiences a rain shadow effect, resulting in drier conditions.
- **Pressure Belts** - Pressure belts are closely related to wind direction and rainfall patterns. Low-pressure areas tend to attract rain-bearing winds, leading to increased precipitation, while high-pressure areas generally do not attract such winds, resulting in less rainfall.

Atmospheric Disturbances

Tropical Cyclone:

A tropical cyclone is a **weather system characterized by low pressure**, originating in the tropics within a single air mass. It can move into temperate waters if the **water temperature is high enough to sustain it**. Tropical cyclones get their energy from the latent heat of condensation. The energy in an average hurricane can be equivalent to more than 10,000 atomic bombs of the size dropped on Nagasaki.

Characteristics of Tropical Cyclones:

Tropical cyclones, which develop in the regions lying between the tropics of Capricorn and Cancer, differ significantly from extratropical or temperate cyclones in their irregularity and diversity. These cyclones vary considerably in shape, size, velocity, and weather conditions, and they play a crucial role in controlling the weather conditions of low latitudes, particularly rainfall regimes. The salient features of tropical cyclones are detailed below:

Size Variation:

- The size of tropical cyclones varies significantly. On average, their diameters range between 80 km and 300 km, but sometimes they can become so small that their diameter is restricted to 50 km or even less.

Velocity:

- Tropical cyclones advance with varying velocities. Weak cyclones move at a speed of about 32 km per hour, while hurricanes can attain velocities of 180 km per hour or more.

Ocean and Land Interaction

- Tropical cyclones become more vigorous and move with very high velocity over the oceans but weaken considerably over land areas. They eventually die out after reaching the interior portions of continents, which is why these cyclones primarily affect coastal areas.

Low Pressure Center:

- The center of a tropical cyclone is characterized by extremely low pressure. The isobars are more or less circular but fewer in number, causing winds to rush toward the center with gale velocity. The air pressure at the center can sometimes be as low as 650 millimetres.

Isobars:

The isobars are generally **circular and close to each other**, resulting in a steep pressure gradient.

Temperature Uniformity:

- Unlike temperate cyclones, tropical cyclones do not have temperature variations in their different parts, as they do not have distinct warm and cold fronts.

Rainfall Distribution:

- There are no different rainfall cells in tropical cyclones, unlike temperate cyclones. Consequently, each part of a tropical cyclone yields rainfall.

Mobility:

- Tropical cyclones are not always mobile. Sometimes, they become stationary over a particular place for several days, causing heavy rainfall, floods, and environmental disasters.

Tracks and Movement:

- The tracks of tropical cyclones vary considerably in different parts. Generally, they move from east to west under the influence of trade winds. The general direction is westerly up to 15° latitude from the equator, poleward between 15°-30° latitudes, and easterly thereafter. These cyclones weaken when they enter subtropical regions.

Seasonal Occurrence:

- Tropical cyclones are confined to particular periods of the year, mainly during the summer season. Their frequency and

Disastrous Impact:

- Tropical cyclones can become disastrous natural hazards due to their high wind speeds (180 to 400 km per hour), high tidal surges, intense rainfall (with recorded values exceeding 2000 mm per day in the Philippines), very low atmospheric pressure causing unusual rises in sea level, and their persistence over several days or about one week over a particular place.

Conditions Favourable for Tropical Cyclone Formation:

- **Sea Surface Temperature** - A large sea surface with temperatures **higher than 27°C**.
- **Coriolis Force** - The Coriolis force, caused by the Earth's rotation, is essential for the cyclonic rotation.
- **Vertical Wind Shear** - Minimal variations in wind speed with altitude are necessary to prevent disruption of the cyclonic structure.
- **Pre-existing Low-pressure Area** - A weak low-pressure area or low-level cyclonic circulation.
- **Upper Atmospheric Divergence** - Divergence in the upper atmosphere facilitates the maintenance of the low-pressure system at the surface.
- **Latent Heat Source** - Ocean waters with temperatures above 27°C provide moisture to feed the storm. **Warm water** should extend **60-70 m below the surface** to support deep convection currents.

Origin of Tropical Cyclones:

The origin of tropical cyclones is not well understood. They generally **develop from small tropical depressions** formed by **easterly waves**, which are areas of lower pressure within the easterly trade winds. When air containing the disturbance is heated by tropical waters with **temperatures of about 27°C or more**, **circular winds begin to blow**. This causes some **warm humid air to be forced upward, leading to condensation and the formation of the storm**. Under ideal conditions, the storm can reach hurricane status, with wind speeds exceeding 118 km/hr, within two to three days.

Early Stage of Development:**Air Uplift and Condensation:**

- In **thunderstorms**, warm and light air is uplifted. At a certain height, **due to normal lapse rate and adiabatic lapse rate, the temperature of the air decreases**. The moisture in the air undergoes **condensation, releasing latent heat**,

which **makes the air warmer and lighter, causing further upliftment**. This cycle continues as long as moisture is supplied, with fresh moisture-laden air being drawn in.

Intensification and Cyclonic Vortex Formation:

- The **thunderstorm intensifies due to the excess moisture over the oceans**, sucking in air at a faster rate. Surrounding air rushes in and undergoes deflection due to the Coriolis force, creating a **cyclonic vortex** (spiralling air column similar to a tornado). Due to centripetal acceleration, the air in the vortex forms a region of calmness called the eye at the center of the cyclone. The inner surface of the vortex forms the eyewall, the most violent region of the cyclone.

Mature Stage of Development:

Convective Cells and Rain Bands:

- In the mature stage, **spiralling winds create multiple convective cells with alternating calm and violent regions**. Regions with **cumulonimbus cloud** formation (rising limbs of the convective cell) are called **rain bands**, below which intense rainfall occurs. Ascending air loses moisture and descends back to the surface through calm regions (descending limbs of convection cell).

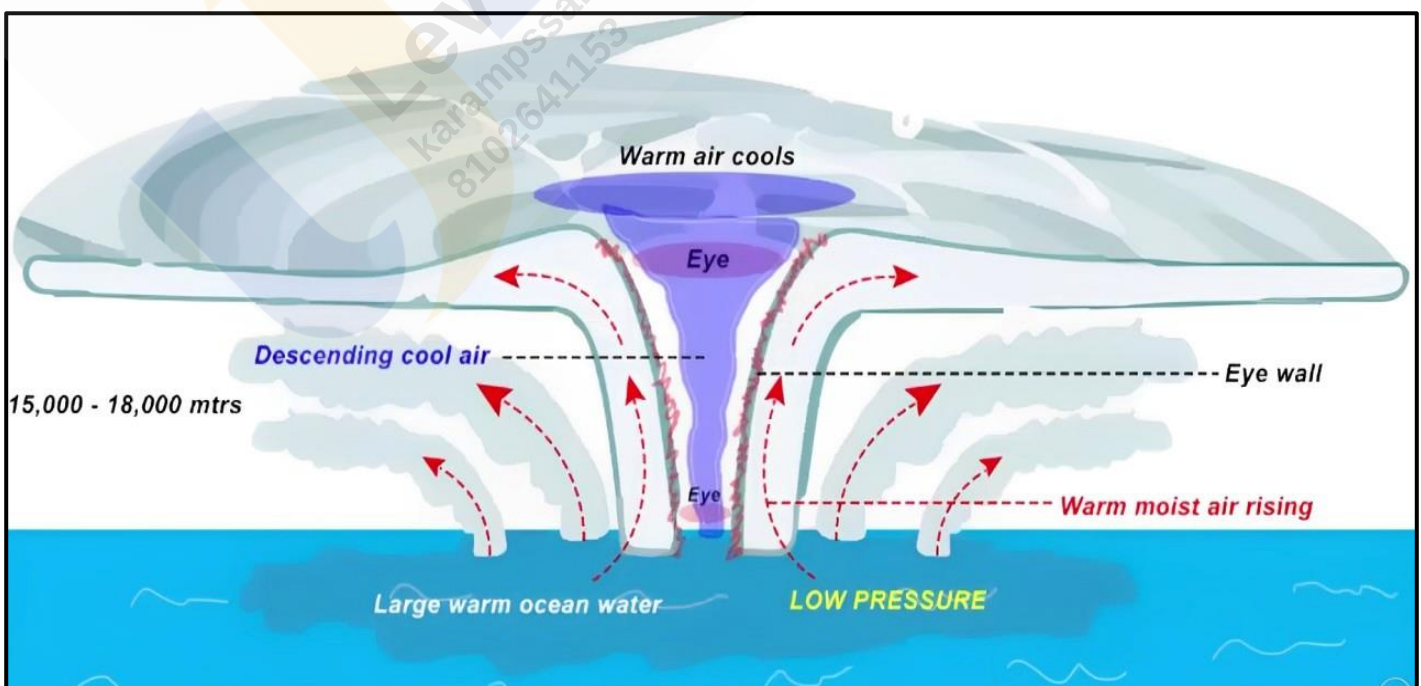
Cloud Formation:

- Cloud formation is densest at the center, decreasing in size towards the periphery**. Rain bands mostly consist of **cumulonimbus clouds**, while the **periphery has nimbostratus and cumulus clouds**. Dense overcast at upper troposphere levels consists of cirrus clouds made up of hexagonal ice crystals. Dry air flowing along the central dense overcast descends at the periphery and eye region.

Structure of a Tropical Cyclone:

Eye:

- A mature tropical cyclone features a **strong spirally circulating wind around the center**, known as the eye. The eye is a roughly **circular area of light winds, clear skies, and fair weather, with little or no precipitation**. It is the region of lowest surface pressure and warmest temperatures aloft, with **temperatures up to 10°C warmer at 12 km altitude than the surrounding environment, but only 0-2°C warmer at the surface**. Eyes range from **8 km to over 200 km in diameter**, with most being 30-60 km across.



Eyewall:

- The eye is surrounded by the eyewall, a **circular ring of deep convection with the highest surface winds** in the tropical cyclone. The eyewall has the **fastest winds and experiences torrential rain**. The air in the eye slowly sinks, while the eyewall has a **net upward flow due to strong convection**. The eye's warm temperatures result from **compressional warming** (adiabatic) of the subsiding air.

Spiral Bands:

- Convection in tropical cyclones is organized into **long, narrow rain bands oriented with the horizontal wind**, called **spiral bands**. These bands have maximum low-level convergence and pronounced upper-level divergence. A direct circulation develops, with **warm, moist air converging at the surface, ascending through the bands, diverging aloft, and descending on both sides of the bands**. Adiabatic warming occurs as air subsides, leading to a **sharp contrast in pressure across the band**, increasing tangential winds and forming the eye and eyewall.

Vertical Structure of a Tropical Cyclone:

- Inflow Layer (0-3 km)** - The lowest layer, responsible for driving the storm.
- Cyclonic Storm Layer (3-7 km)** - The main cyclonic activity takes place here.
- Outflow Layer (Above 7 km)** - The outflow layer extends above 7 km, with maximum outflow at 12 km and above. The air movement is anticyclonic.

Nomenclature of Tropical Cyclones:

- Hurricanes** - Western North Atlantic, Central and Eastern North Pacific, Caribbean Sea, and Gulf of Mexico.
- Typhoons or Taiphoo** - Western North Pacific, China, and Japan.
- Cyclone or Chakravaat** - Bay of Bengal and Arabian Sea
- Baguio** - Philippines.
- Willi Willies** - Australia.

Naming of Cyclones:

One of the earliest attempts at naming cyclones was by Clement Wragge, a Queensland Government meteorologist in Australia. Between 1887 and 1907, he used names from Greek mythology and female names for weather systems. In the mid-1900s, it became common to use feminine names for cyclones. This practice was influenced by the tradition of naming ships as female. Recognizing the need for a more organized system, meteorologists started using alphabetical lists of names for cyclones. This made it easier to identify and track storms.

Today, the naming of cyclones is managed by various regional bodies under the guidelines of the **World Meteorological Organization (WMO)**. Cyclones are named to provide clarity in communication and to avoid confusion when multiple systems are active simultaneously.

Regional Specialized Meteorological Centers (RSMCs) and **Tropical Cyclone Warning Centres (TCWCs)** play crucial roles in monitoring and forecasting tropical cyclones. There are **six RSMCs** and **five TCWCs** worldwide, each responsible for specific ocean basins. **These centers are responsible for tracking storms, issuing warnings, and providing information on cyclone development, trajectory, and intensity.**

Regional Specialized Meteorological Centers (RSMCs):

- RSMC Tokyo (Japan Meteorological Agency)
- RSMC New Delhi (India Meteorological Department)

- RSMC Honolulu (Central Pacific Hurricane Center)
- RSMC Miami (National Hurricane Center)
- RSMC La Réunion (Meteo-France)
- RSMC Nadi (Fiji Meteorological Service)

Tropical Cyclone Warning Centres (TCWCs):

- TCWC Melbourne (Bureau of Meteorology, Australia)
- TCWC Perth (Bureau of Meteorology, Australia)
- TCWC Darwin (Bureau of Meteorology, Australia)
- TCWC Jakarta (Badan Meteorologi, Klimatologi, dan Geofisika, Indonesia)
- TCWC Port Moresby (Papua New Guinea National Weather Service)

Guiding Principles for Cyclone Naming Process:

- **Neutrality** - Proposed names should neutralize politics, religious beliefs, cultures, and gender.
- **Sensitivity** - Names should avoid hurting the sentiments of any global population group.
- **Respectful Nature** - Names should not be rude or cruel and should be short, easy to pronounce, and non-offensive.
- **Maximum Length** - Cyclone names are limited to eight letters.

Role of the India Meteorological Department (IMD):

IMD, one of the six RSMCs in the world, is mandated to issue advisories and name tropical cyclones in the north Indian Ocean region. The advisories are issued to **13 member countries** under the WMO/ESCAP Panel including - **Bangladesh, India, Iran, Maldives, Myanmar, Oman, Pakistan, Qatar, Saudi Arabia, Sri Lanka, Thailand, United Arab Emirates, and Yemen.**

Benefits of Naming Tropical Cyclones:

- **Identify each individual cyclone** - Ensures each cyclone is easily distinguishable.
- **Create awareness of its development** - Enhances public awareness regarding the presence and progress of cyclones.
- **Remove confusion** - Avoids confusion in case of simultaneous occurrence of tropical cyclones in a region.
- **Remember a tropical cyclone easily** - Simplifies the process of remembering cyclones.
- **Disseminate warnings rapidly and effectively** - Facilitates quick and broad dissemination of warnings to the public.

Detection Parameters of Cyclones:

- **Atmospheric Pressure** - Cyclones feature a significant drop in atmospheric pressure at their centers, with lower pressures indicating stronger storms. The steep pressure gradient leads to strong winds.
- **Wind Speed and Direction** - High-speed surface winds, measured by anemometers and satellites, are key indicators of cyclone strength. Low vertical wind shear aids in cyclone formation and intensification.
- **Temperature** - Warm Sea surface temperatures (above 26.5°C) provide energy for cyclone formation. A greater temperature gradient between the sea surface and the upper atmosphere promotes development.
- **Humidity** - High humidity levels in the mid and lower troposphere are crucial for cyclone development. Satellite sensors measure atmospheric moisture, detecting water vapor content necessary for intensification.
- **Cloud Patterns and Precipitation** - Distinct cloud patterns and heavy rainfall are tracked using satellite imagery and radar systems, helping to understand cyclone structure and impacts.

Temperate Cyclone:

Temperate cyclones, also known as **extra-tropical cyclones**, **mid-latitude cyclones**, **frontal cyclones**, or **wave cyclones**, are **large-scale low-pressure weather systems** that occur in the temperate zones of the Earth, primarily between 30° and 60° latitude in both hemispheres. They are characterized by rotating winds around a central low-pressure area and are associated with fronts, where contrasting air masses meet. Temperate cyclones play a **crucial role in the distribution of heat and moisture across the globe**, influencing weather patterns and climatic conditions.

Origin and Development of Temperate Cyclones:

Polar Front Theory:

According to the polar front theory, temperate cyclones form at the **boundary between warm, humid air masses from the tropics and cold, dry air masses from the poles**, known as the '**polar front**.' This boundary acts as a surface of discontinuity, where the contrasting air masses interact to create storm systems.

Formation of Polar Front:

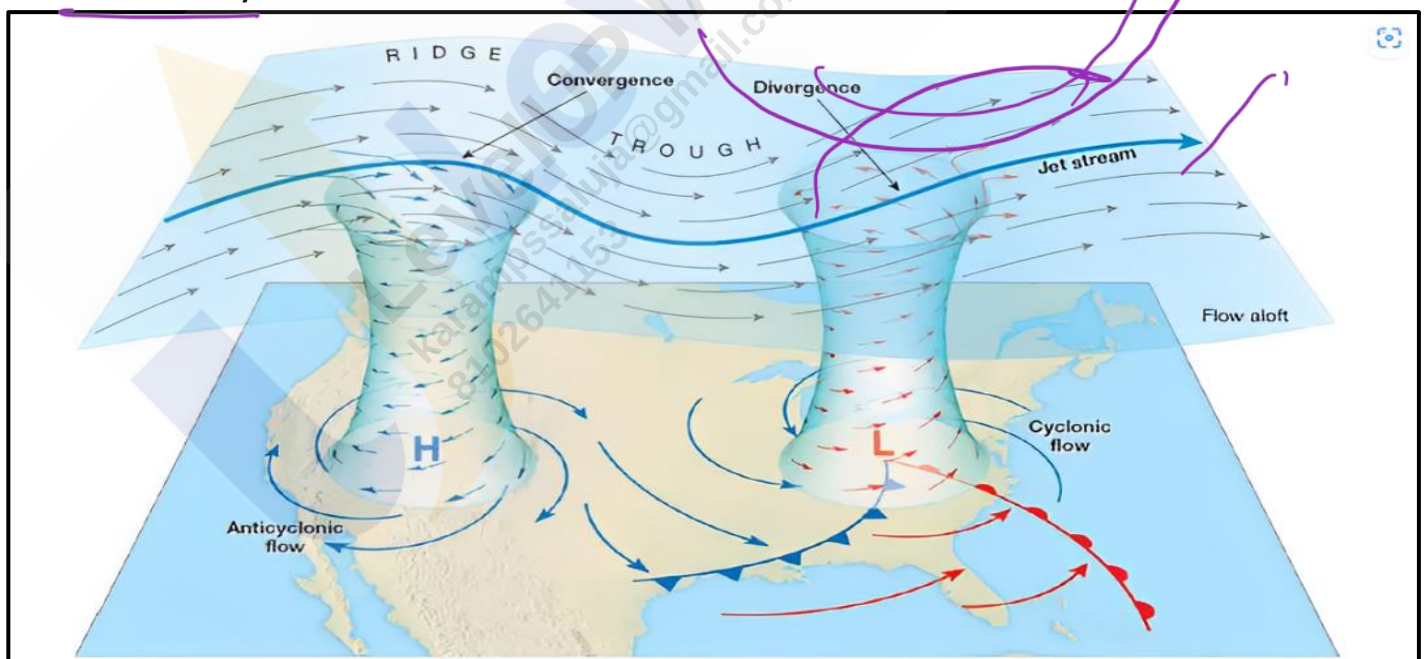
- The polar front is formed where **warm, humid tropical air meets cold, dry polar air**. This interaction creates a zone of instability, setting the stage for cyclone formation.

Cyclogenesis:

- The cold air pushes the warm air upwards from underneath, creating a void due to decreased pressure. Surrounding air rushes in to fill this void, and the Earth's rotation causes a cyclonic circulation. The cyclone advances with the westerlies (jet streams).

Warm and Cold Air Movements:

- In the northern hemisphere, warm air moves from the south and cold air from the north. When pressure drops along the front, the warm air moves northwards, and the cold air moves southwards, setting in motion an **anticlockwise cyclonic circulation due to the Coriolis force**.



Development Stages:

Initial Stage:

- A well-developed extra-tropical cyclone has distinct warm and cold fronts. Warm air is wedged between advancing cold air masses.

Mature Stage:

- The warm air glides over the cold air, forming a sequence of clouds and causing precipitation. The cold front approaches the warm air from behind, pushing it upwards, forming cumulus clouds.

Occluded Stage:

- The **cold front moves faster than the warm front and eventually overtakes it**, lifting the warm air entirely and forming an **occluded front**. The cyclone then begins to **dissipate**.

Cyclone Lifespan:

- Individual frontal cyclones typically last 3 to 10 days and move in a generally west to east direction, influenced by the polar jet stream.

**Characteristics of Temperate Cyclones:****Structure:**

- A typical mature mid-latitude or temperate cyclone is an expansive low-pressure system with a **diameter of approximately 1600 kilometers (1000 miles)**. These cyclones form vast cells of low-pressure air, with **central ground-level pressure typically ranging between 990 and 1000 millibars**. Generally, the system exhibits an **oval shape, with its long axis trending in a northeast-southwest direction**. A distinct pressure trough often extends south-westerly from the cyclone's center.

Formation of Fronts:

- In the **Northern Hemisphere**, temperate cyclones exhibit a **converging counterclockwise circulation pattern**. This wind flow brings together cooler air masses from the north and warmer air masses from the south. The convergence of these differing air masses typically creates two distinct fronts:
- Cold Front** - This front extends southwest from the cyclone's center and aligns along the pressure trough.
- Warm Front** - This front extends eastward from the cyclone's center and runs along another, typically weaker, pressure trough.

Sectors of the Cyclone:

- Cool Sector** - Located north and west of the cyclone's center, this sector is where the cold air mass is in contact with the ground. At the surface, the cool sector is larger.
- Warm Sector** - Situated to the south and east of the center, this sector has the warm air mass in contact with the ground. Although smaller at the surface, the warm sector is more extensive aloft due to the "leaning" nature of both fronts. The cold front slopes upward toward the northwest, and the warm front slopes upward toward the northeast.

Clouds and Precipitation:

- Cloud formation and precipitation in mid-latitude cyclones occur in **zones where air is rising and cooling adiabatically**. Since **warm air rises along both fronts, two zones of cloudiness and precipitation** typically form, overlapping around the cyclone's center and extending outward along the fronts.
- Cold Front** - Along and immediately behind the cold front's ground-level position (the steeper of the two fronts), **cumuliform clouds** often form, resulting in showery precipitation.
- Warm Front** - The more gradual slope of the warm front leads to the development of **extensive horizontally oriented clouds**, which may produce **widespread, prolonged, low-intensity precipitation**. **Most precipitation originates from the warm air rising above the fronts** and falls through the front to reach the ground in the cool sector.

Weather Patterns:

- The precipitation pattern does not imply that the entire cool sector experiences unsettled weather while the warm sector remains clear. While **most frontal precipitation falls within the cool sector**, the general area north, northwest, and **west of the cyclone's center often becomes cloudless once the cold front passes**. Thus, **much of the cool sector is characterized by clear, cold, stable air**.
- Conversely, the **warm sector's air tends to be moist and unstable**. This instability, coupled with surface-wind convergence, can lead to sporadic thunderstorms. Occasionally, **one or more squall lines of intense thunderstorms** may develop in the warm sector ahead of the cold front.

Seasonal Occurrence of Temperate Cyclones:

Temperate cyclones occur throughout the year but exhibit seasonal variations in their frequency and intensity:

Winter:

- Temperate cyclones are most frequent and intense during the winter months. The larger temperature contrasts between polar and tropical air masses during this season lead to stronger cyclogenesis.

Spring and Autumn:

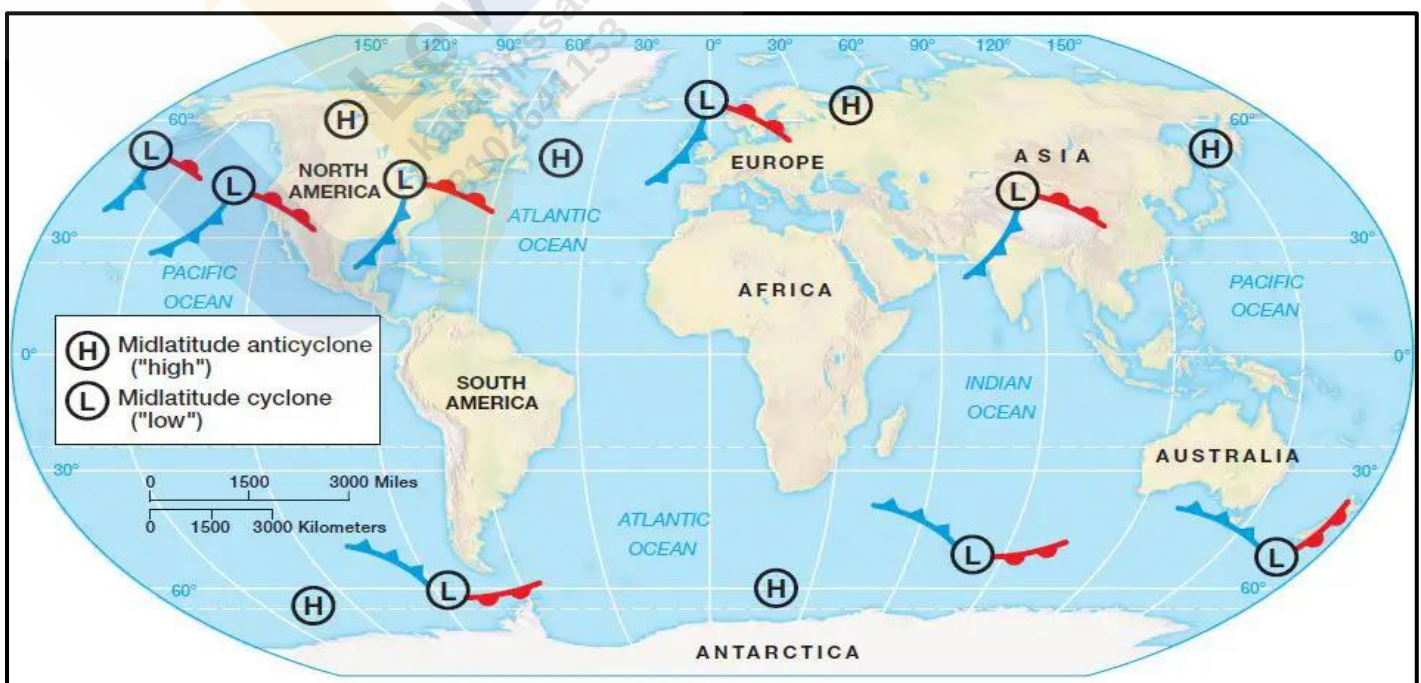
- These seasons also see a significant number of temperate cyclones, as transitional weather patterns promote cyclogenesis.

Summer:

- Temperate cyclones are less common and weaker in the summer due to reduced temperature gradients.

Distribution of Temperate Cyclones:

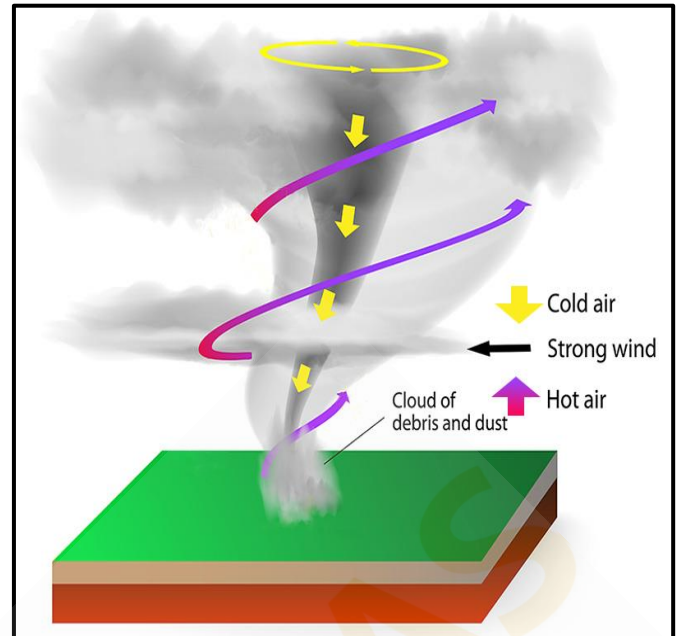
- USA and Canada** - Extend over Sierra Nevada, Colorado, Eastern Canadian Rockies, and the Great Lakes region.
- Northern Europe and Russia** - From Iceland to the Barents Sea, continuing over Russia and Siberia.
- Baltic Sea** - Experiences winter storms.
- Mediterranean Basin** - Extends to Russia and even India in winters (western disturbances).
- Antarctic Frontal Zone** - Regular occurrence of temperate cyclones.



Tornado:

A tornado is a **violently rotating column of air extending from a thunderstorm to the ground**. Characterized by its funnel shape and rapid rotation, a tornado is **one of nature's most destructive weather phenomena**. It forms due to specific meteorological conditions and can cause significant damage through both high-velocity winds and air pressure reductions. The destructive power of a tornado is due to the high wind speeds it can generate, which can reach up to 300 miles per hour. These high winds can lift heavy objects off the ground and carry them long distances.

It is often referred to as a **twister, whirlwind, or cyclone**, although the word cyclone is used in meteorology to name a weather system with a low-pressure area in the center around which, from an observer looking down toward the surface of the Earth, winds blow **counterclockwise in the Northern Hemisphere** and **clockwise in the Southern Hemisphere**.



Origin of a Tornado:

Tornadoes are formed **when warm, moist air meets cold, dry air**. This creates an unstable atmosphere, which can lead to the formation of **thunderstorms**. Under the right conditions, a thunderstorm can produce a **rotating column of air** called a **mesocyclone**. If the mesocyclone reaches the ground, it becomes a tornado.

Several factors contribute to the formation of a tornado:

- **Wind Shear** - The change in wind speed and direction with height, is crucial for tornado formation. This creates a horizontal spinning effect within a storm cell.
- **Lift** - Rising air from the surface is needed to lift the horizontal spinning air column into a vertical orientation.
- **Instability** - Atmospheric instability, where warmer, moist air at the surface rises and cools, contributes to storm development.
- **Moisture** - Moisture in the air is essential for cloud formation and subsequent thunderstorm development.

Distribution of Tornadoes:

Global Distribution:

United States:

- Tornadoes are most frequent in the United States, particularly in "**Tornado Alley**," which includes parts of Texas, Oklahoma, Kansas, and Nebraska. The U.S. experiences more tornadoes than any other country.

Canada:

- Tornadoes occur in Canada, with the highest frequency in the provinces of Alberta, Saskatchewan, and Manitoba.

Australia:

- Tornadoes are less common in Australia but do occur, particularly in the southeastern regions and occasionally in Queensland.

Europe:

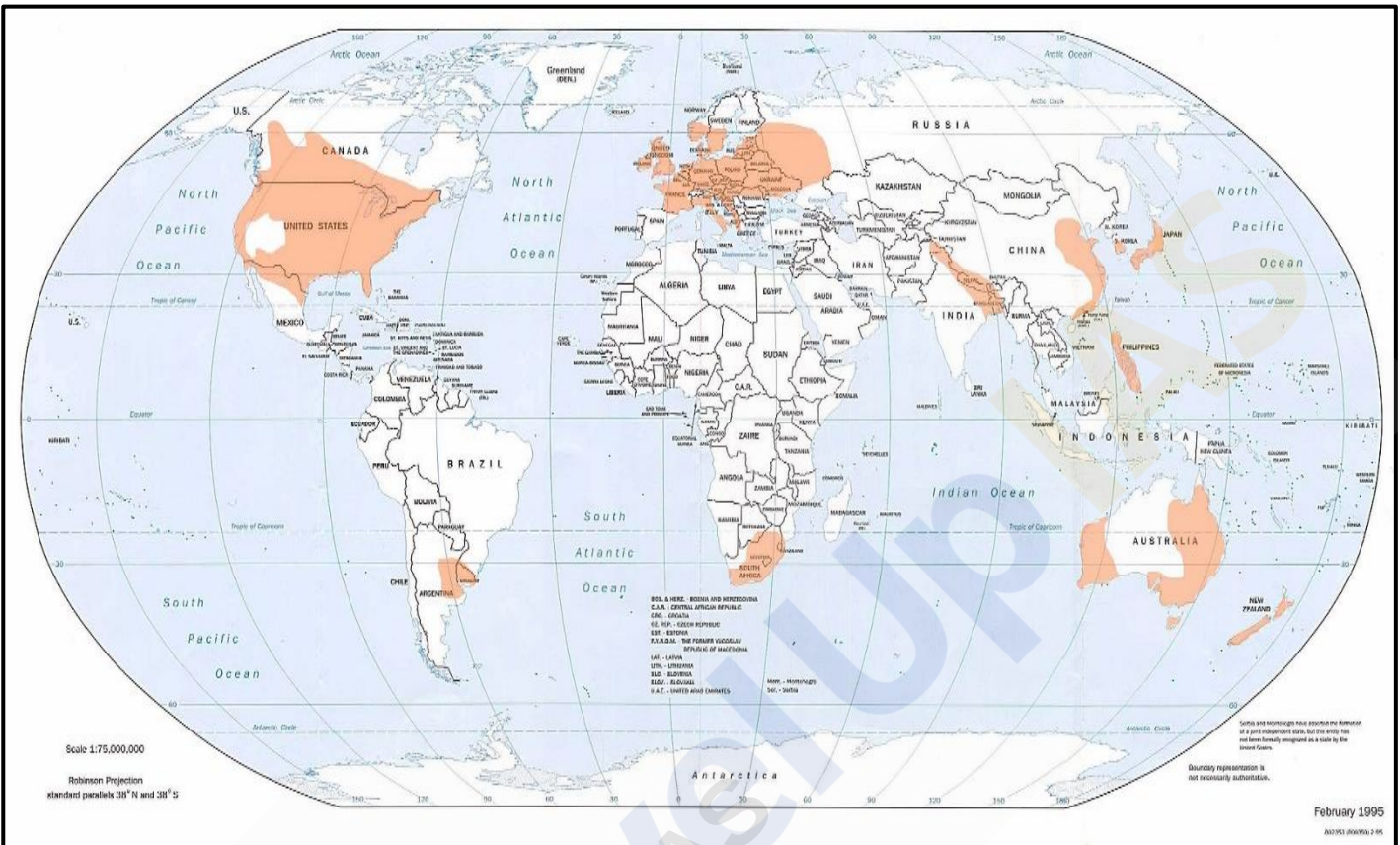
- Tornadoes are relatively rare in Europe, but they do occur, with the highest frequency in countries like the United Kingdom, the Netherlands, and Germany.

South America:

- Tornadoes are rare but do occur in parts of Argentina, Brazil, and Paraguay.

Asia:

- Tornadoes are infrequent in Asia, but they have been reported in countries like **Bangladesh**, India, and Japan, especially during severe thunderstorms or cyclonic conditions.



Seasonal Variation:

Peak Seasons:

- Tornadoes are most common in spring and fall when wind shear and atmospheric instability are most pronounced. They are least common in winter.

Time of Day:

- Tornado occurrence is influenced by solar heating, which affects the development of thunderstorms.

Factors Influencing Distribution:

- Geography** - Tornadoes are more common in areas with large flat plains and where warm, moist air meets cooler, drier air.
- Seasonal Patterns** - Tornadoes often occur in spring and early summer when conditions for severe thunderstorms are more prevalent.
- Climate** - Regions with temperate climates and significant seasonal temperature contrasts are more likely to experience tornadoes.

Tornadoes, as well as cyclones both, occur in India. However, unlike cyclones, the frequency of tornado outbreaks is very low. **Cyclones originate in the Bay of Bengal region as well as in the Arabian Sea region** whereas **Tornadoes of weak strength occur in the north-western and north-eastern region** of the country causing significant damage to man and material.